AiRAiDE

Ai Grid

Presents

10 weeks

AGENTIC AI MASTER CLASS

Hands-on GPU-powered training for building, deploying, and serving real-world AI models.



ENROLL NOW!

TAKE THE NEXT STEP IN YOUR AI JOURNEY.

Register Now!

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Overview:



Join our 10-week intensive intermediate AI Masterclass designed for professionals, researchers, and innovators eager to master the full AI lifecycle. From data engineering to advanced deep learning architectures like Transformer architecture, training, quantization, and fine-tuning small Language models, generative AI, and simple AI agents. This program blends theory, hands-on labs, and real-world case studies on GPU clusters.

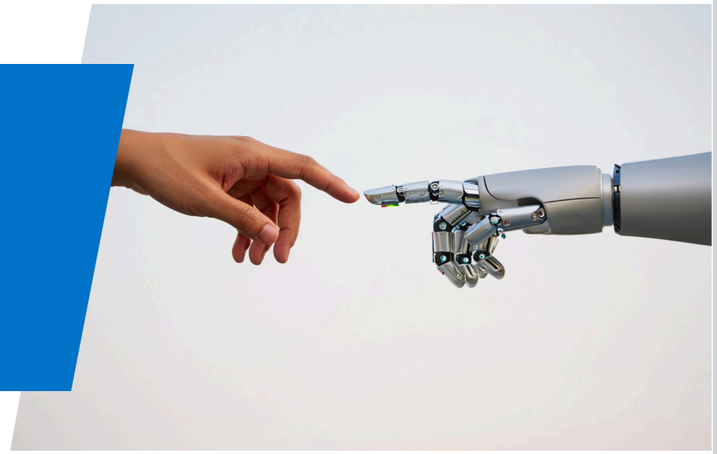


Immersion Lab Tools



Real-world case studies

Week 1 & 2: DATA ENGINEERING



Learning Focus: Understanding the data engineering lifecycle, building data pipelines, and learning how modern systems move data from sources to storage to analytics.

Week 1 establishes the foundational concepts and practical skills required to operate as a data engineer in modern technology environments.

The week begins with an overview of Data Engineering as a discipline. Learners explore how modern companies move data across systems and the core components of a data pipeline. Key concepts such as data sources (APIs, Databases, Logs, IoT), ETL (extract, transform, Load) vs ELT (Extract, Load, Transform), and basic data architecture are introduced.

A major focus is placed on building data pipelines, which is the backbone of any data engineering workflow. Students learn how data flows through modern systems, how to design structured ETL pipelines, and how to extract data from APIs. Data formats including CSV, JSON, and Parquet are covered, alongside pipeline orchestration and workflow scheduling. Students learn to transform data programmatically and automate data ingestion tasks using scripts.

The week also emphasizes modern data architecture, how large-scale companies store and manage massive volumes of data. Learners study the differences between:

- Data Warehouses
- Data Lakes
- ETL vs ELT Pipelines
- Batch vs Streaming Processing.

Hands-on Labs:

- Download and load a CSV dataset using Python and Pandas libraries. Clean basic errors and store the data into SQLite/PostgreSQL.
- Extract data from a public API. Transform the JSON data. Store the results in a database.
- Take raw dataset, Transform it into analytical tables, Run analytical queries.

Outcome:

By the end of Week 1, learners understand the

- Describe how data flows through modern systems. Build a basic ingestion pipeline.
- Transform raw data into analytics-ready datasets.
- Extract data from APIs. Transform data programmatically. Automate data workflows.

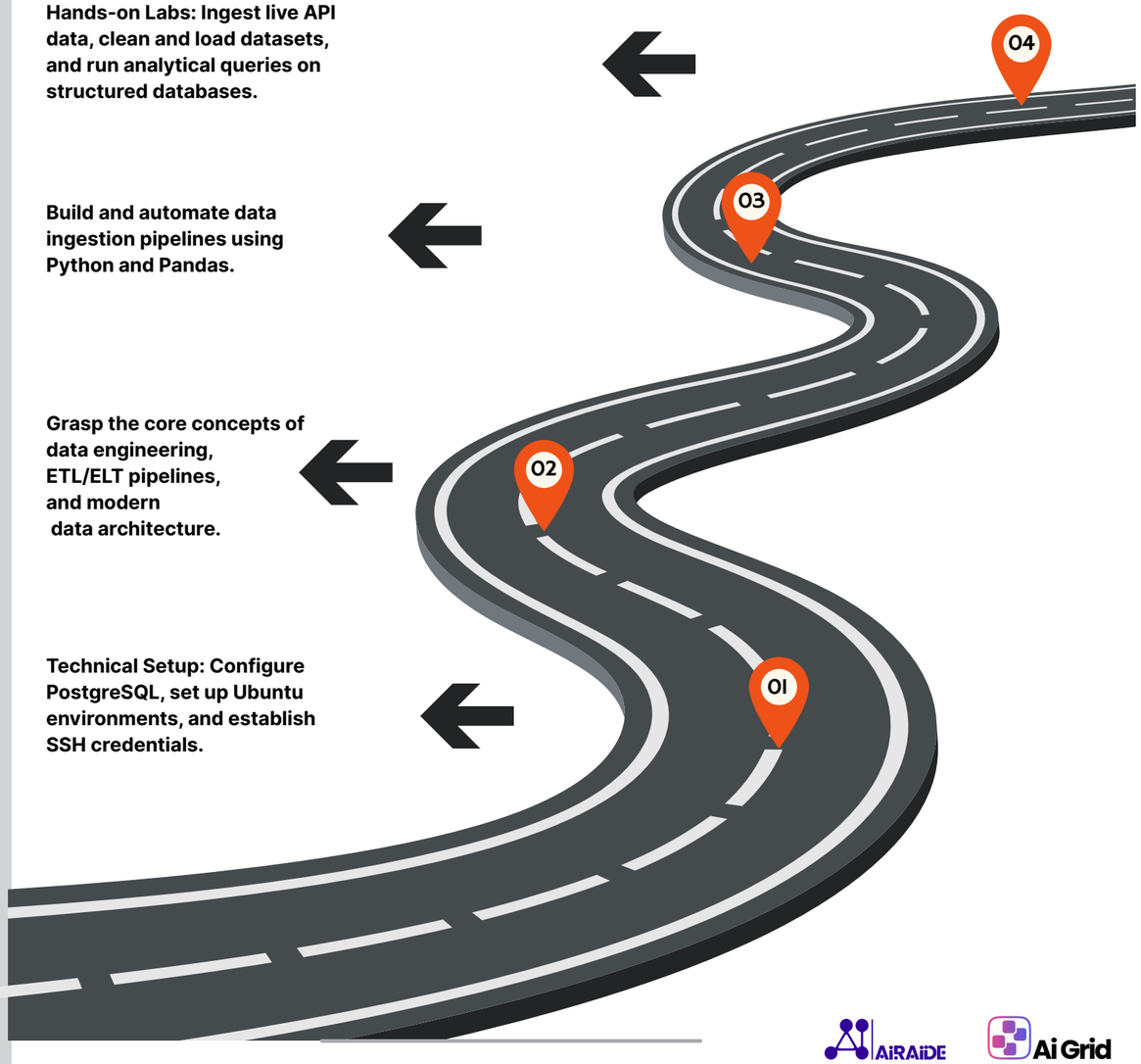
Week 1 & 2: DATA ENGINEERING

Hands-on Labs: Ingest live API data, clean and load datasets, and run analytical queries on structured databases.

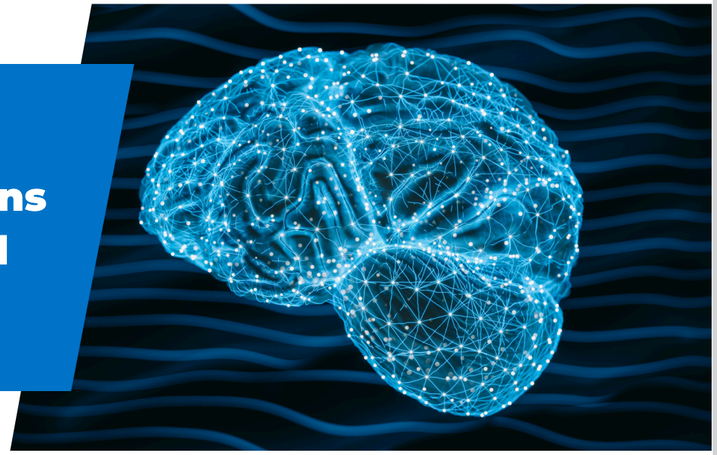
Build and automate data ingestion pipelines using Python and Pandas.

Grasp the core concepts of data engineering, ETL/ELT pipelines, and modern data architecture.

Technical Setup: Configure PostgreSQL, set up Ubuntu environments, and establish SSH credentials.



Week 3 & 4: Deep Learning Foundations Neural Network and Transformers.



Week 2 builds on Data fundamentals and dives into deep learning architectures that power modern AI systems.

The focus is on traditional neural networks and deeper architectures, including transformer-based models. Students explore loss functions, activation functions and how deep neural networks automatically learn hierarchical feature representations from raw data.

A significant component of the week centers on the Transformer architecture, introduced in the landmark paper “Attention Is All You Need”. Learners examine:

- Self-attention mechanisms
- Multi-head attention
- Positional embeddings

In addition, learners are introduced to modern model packaging and deployment formats, including GGUF and Safetensors, alongside common training paradigms such as pre-training (PT) and instruction tuning (IT). The course also explains precision data formats used in model storage and inference, including INT8, FP16, and FP32, helping students understand how model representation impacts memory usage, performance, and deployment efficiency.

Learners also explore open-source representations of transformer architectures and the open-source libraries used to train and experiment with transformer models. Through this exploration, students examine how dataset selection, tooling choices, and model design decisions influence fairness, transparency, and reliability in language models.

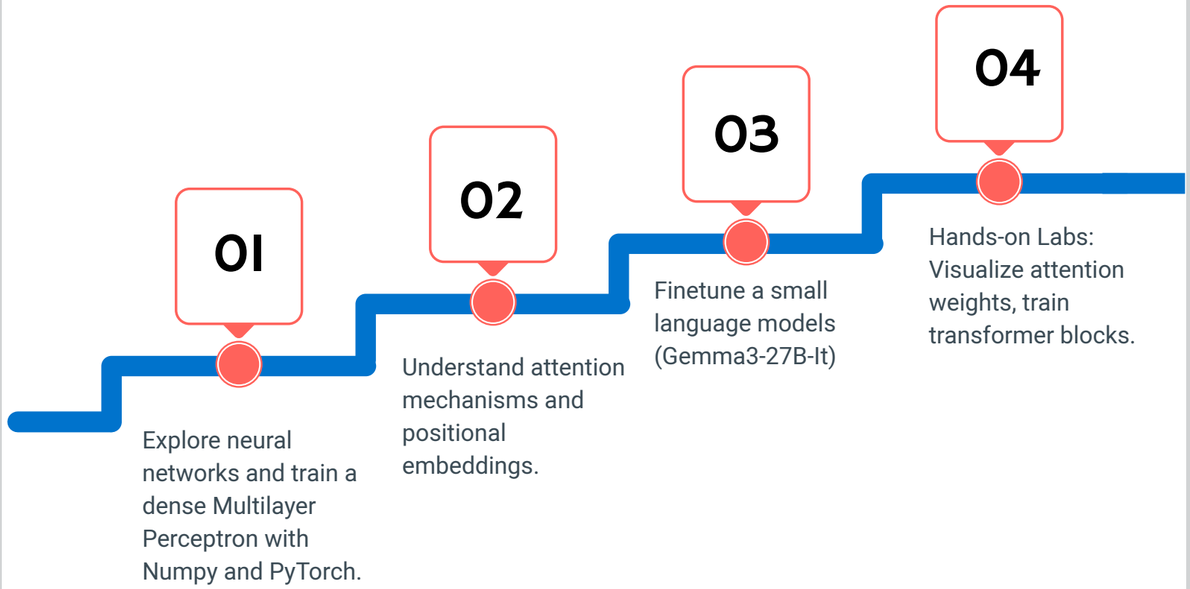
Hands-on Labs:

- Visualize attention weights in transformer models.
- Implement a transformer from scratch in Python
- Train your own Language model on opensource data
- Finetune small Language model (Gemma3-27B-it)

Outcome:

By the end of Week 2, learners would have a mental model for how neural networks learn. They would understand the components of transformers and how they work. They can build and train a multilayer perceptron and transformers. They can quantise and finetune language models.

Week 3 & 4: Deep Learning Foundations Neural Network and Transformers.



Week 5 & 6: Large Language Models, Agents



Week 3 explores advanced generative AI systems, focusing on Large Language Models (LLMs), intelligent agents, and evaluation frameworks. Students begin with prompt engineering, learning how to craft effective prompts using techniques such as:

- Zero-shot prompting
- Few-shot prompting
- Chain-of-thought reasoning

The week introduces the architecture of modern AI agent systems. Students learn how agents operate through an intelligence layer that performs reasoning and planning, and a memory layer that stores and retrieves contextual information. The course explains how agents interact with external tools through structured tool calls and API integrations to complete tasks.

Core concepts such as tokenization and embedding models are explored to show how text is converted into numerical representations that language models can process. Learners examine how embeddings enable semantic search, memory retrieval, and contextual grounding in agent workflows.

Students also explore how developers interact with AI systems through APIs. Practical examples demonstrate common capabilities such as text generation, audio transcription, and language translation, illustrating how these services are combined to build real-world AI applications.

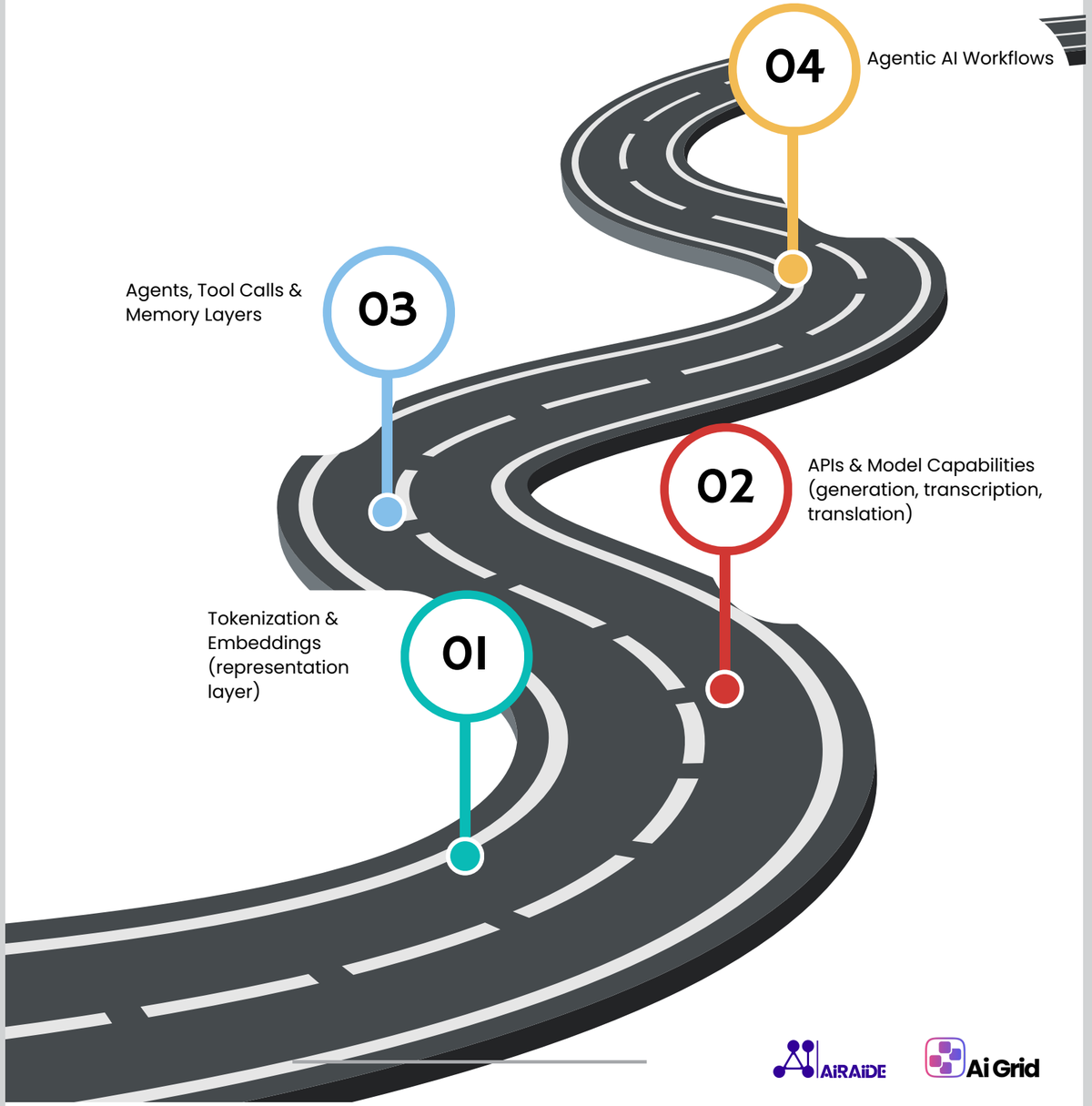
Hands-on Labs:

- Build simple AI agents capable of calling external tools and APIs.
- Implement agentic workflows that combine text generation, embeddings, and memory retrieval.
- Experiment with audio transcription and translation pipelines using AI APIs.

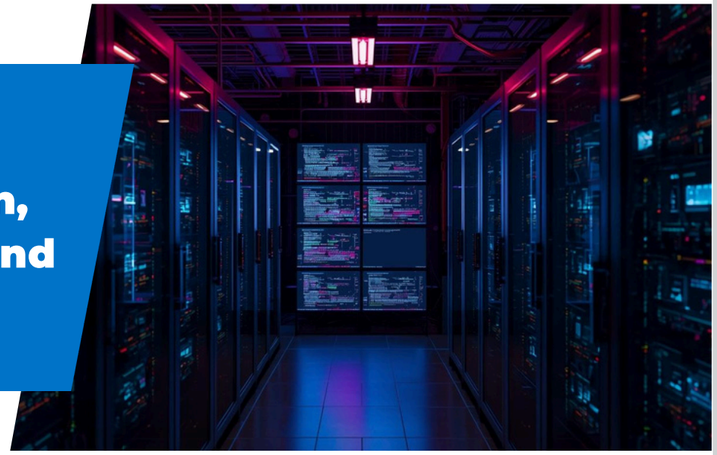
Outcome:

By the end of Week 3, learners understand how agentic AI systems are structured and can implement basic AI agents that use tokenization, embeddings, API calls, and tool integrations to perform autonomous workflows.

Week 5 & 6: Large Language Models, Agents



Weeks 7 - 10: CUDA Kernels with Python, GPU Architecture, and AI Inference



Week 4 focuses on building production-ready AI systems by understanding the hardware and software layers that power modern AI workloads. Students begin with a conceptual and visual exploration of modern GPU architecture and how specialized hardware enables high-performance computing for machine learning and numerical workloads.

Learners examine GPU design principles, including chip layout, high-bandwidth memory, Streaming Multiprocessors (SMs), warp scheduling, and Tensor Cores. The course explains the GPU memory hierarchy—from registers and shared memory to global memory—and how these components interact to enable massive parallelism and high throughput. This knowledge enables students to grow as hardware-aware programmers. Students analyze performance behavior using architectural diagrams and visual profiling tools.

The week introduces CUDA programming using Python, allowing students to write and launch custom GPU kernels. Learners accelerate numerical Python programs by executing computations across thousands of parallel GPU threads.

Students then explore AI inference systems and how trained models from earlier lessons are deployed efficiently on GPU infrastructure. The course examines model serialization formats and the considerations involved in selecting model formats for deployment, including portability, efficiency, and security. Learners also study modern inference engines and serving frameworks used to run large models in production environments (vLLMs, SGLang).

Hands-on Labs:

- Distributed serving of AI coding agents with open-weight models like gpt-oss-120B.
- Implement vector addition and matrix operations as CUDA kernels using Python and Numba.
- Deploy and evaluate GPU-accelerated AI inference pipelines.

Outcome:

By the end of Week 4, learners understand how GPU hardware enables large-scale AI computation and can write CUDA kernels in Python, analyze GPU performance behavior, and deploy optimized AI inference workloads on GPU systems.

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Weeks 7-10: CUDA Kernels with Python, GPU Architecture, and AI Inference

01	02	03	04
Hopper architecture, SMs, Tensor Cores, HBM, memory hierarchy, profiling with Nsight.	CUDA Kernels with Python: Thread hierarchy, grids/blocks, shared memory, host ↔ device transfers.	GPU Performance Optimization: Parallel execution, memory coalescing, tiling, analyzing bottlenecks.	AI Inference Systems: Model formats, inference engines (vLLM / SGLang), distributed serving, latency & throughput evaluation.

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Ideal For:

Data Scientists & ML Engineers

AI Researchers & Developers

Tech Leaders & Innovators

PREREQUISITES

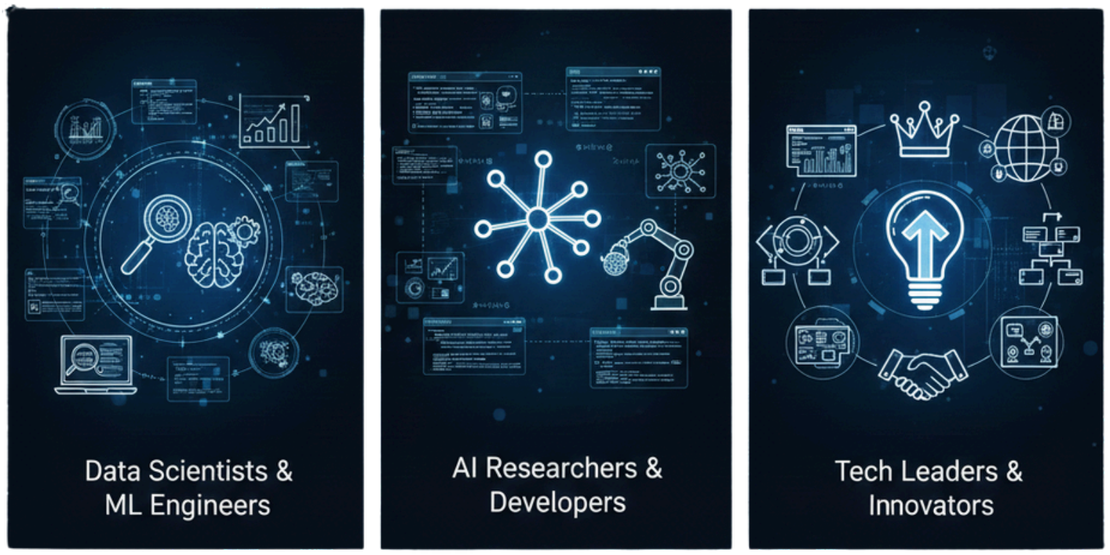
Intermediate Python programming

Basic linear algebra and matrix operations

Familiarity with APIs and JSON data formats

Basic command line usage (Linux/Mac/WSL)

Introductory machine learning concepts



Why Join
This Masterclass

Joining this AI Masterclass offers more than theoretical knowledge; the training provides a comprehensive, practical, and ethically grounded learning experience designed to prepare you for real-world impact

Systems-Level AI Engineering: From Models to Infrastructure

This masterclass goes beyond surface-level AI usage to teach how modern AI systems actually work across the full stack. Participants explore transformer architectures, agentic workflows, GPU computing, and AI inference systems, building an understanding that spans from tokenization and embeddings to CUDA kernels and GPU acceleration. Learners gain the ability to design and optimize production-ready AI solutions.

Hands-On Development with Modern AI Tools on a 4 x RTX5090 Cluster

Each session combines conceptual learning with practical implementation using the same tools and frameworks used in industry. Participants work with open-source transformer libraries, build agentic workflows with tool calls and APIs, write custom CUDA kernels using Python, and deploy AI workloads across 4 GPUS. Through structured labs and guided experiments, learners gain experience of distributed computing, model sharding, and AI serving. This knowledge is foundational to building real-world AI applications.

Production-Oriented Perspective on AI Systems

This masterclass emphasizes how AI systems operate in real-world environments where performance, scalability, and reliability matter. Learners examine GPU architectures, model serialization formats, inference engines, and distributed serving strategies used to run modern AI models at scale. By understanding deployment pipelines, performance bottlenecks, and system-level optimization, participants gain the mindset and technical knowledge required to build AI solutions that are not only intelligent but also efficient and deployable in production environments.

Expert-Led Sessions Combining Theory and Application

Upon successful completion, participants receive a professional certificate recognizing their mastery of modern AI tools and methodologies. The credential demonstrates expertise in deep learning, generative AI, and deployment practices. This enhances career opportunities and professional credibility in the rapidly evolving AI field

Facilitators Bio

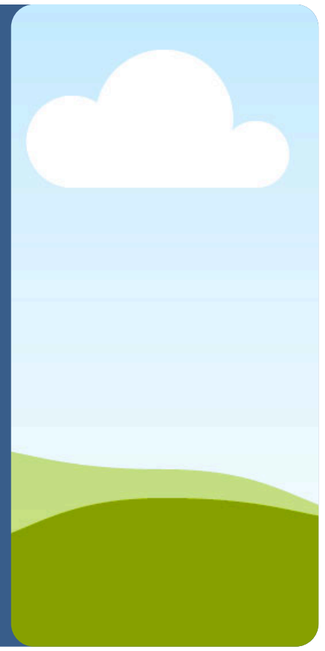
EMMANUEL AKINGBADE



Field: Data Engineering

Emmanuel is a technologist focused on data, automation, and intelligent systems in Industry 4.0 environments. His work explores how software and physical processes connect in real-world systems, with a long-term focus on data and systems-level engineering for resilient industrial infrastructure.

Andrew Ng



Field: High-Performance Computing (HPC) Engineer

Andrew Ng is a leading AI researcher, educator, and entrepreneur known for making AI education accessible worldwide. He co-founded Coursera and previously led AI research at Baidu. He was also a founding member of Google Brain. Ng has played a key role in shaping modern machine learning and AI infrastructure used in data

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Joseph Babajide



Field: AI Engineering

Joseph is a biomedical engineer working with AI agents and large language models in enterprise systems such as Odoo. His work combines artificial intelligence, simulation, and digital communication strategy to design practical technology solutions that support innovation, automation, and business growth.

Victor Ogundele



Field: AI Engineering

Victor bridges AI research and engineering, specializing in computer vision, embedded systems, and robotics for healthcare applications. His work includes healthcare record digitization using computer vision, low-cost IoT patient monitoring systems, and research on AI-driven robotics for medical services.

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